

Name



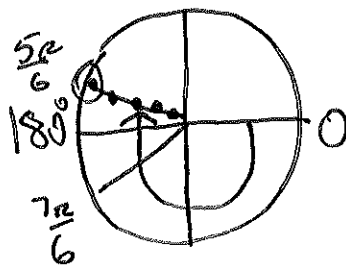
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Math 104 Spring 2025 Final Exam

- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

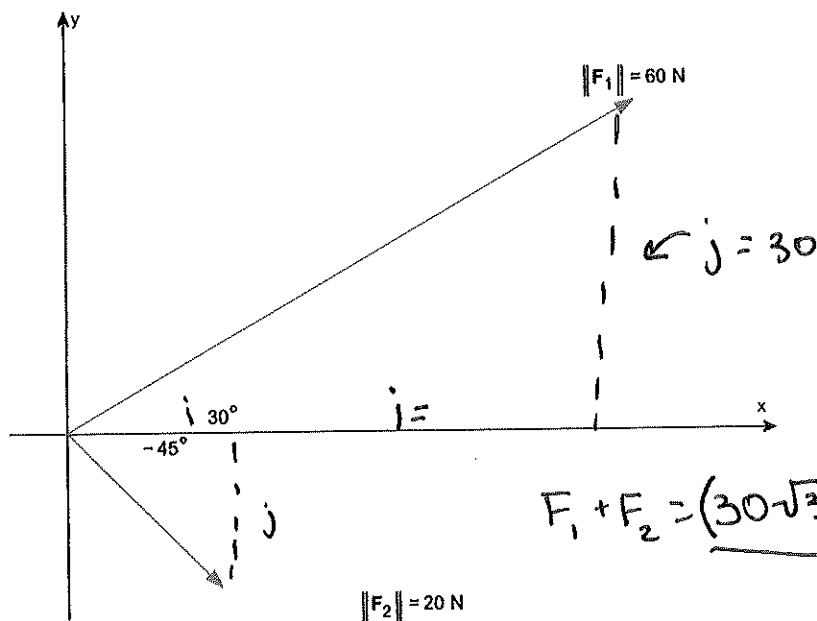
Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$

Soh cah toa



$$F_1 + F_2 = (30\sqrt{3} + 10\sqrt{2})i + (-10\sqrt{2} + 30)j$$

$$F_1) \sin 30^\circ = \frac{j}{60} \rightarrow \frac{1}{2} = \frac{j}{60} \rightarrow \frac{60}{2} = \frac{j}{2} \rightarrow \boxed{j = 30}$$

$$\cos 30^\circ = \frac{i}{60} \rightarrow \frac{\sqrt{3}}{2} = \frac{i}{60} \rightarrow 2i = 60\sqrt{3} \rightarrow \boxed{i = 30\sqrt{3}}$$

$$F_1 = 30\sqrt{3}i + 30j$$

$$F_2) \sin -45^\circ = \frac{j}{20} \rightarrow \frac{-\sqrt{2}}{2} = \frac{j}{20} \rightarrow \frac{2j}{2} = \frac{-20\sqrt{2}}{2} \rightarrow \boxed{j = -10\sqrt{2}}$$

$$\cos -45^\circ = \frac{i}{20} \rightarrow \frac{\sqrt{2}}{2} = \frac{i}{20} \rightarrow \frac{2i}{2} = \frac{20\sqrt{2}}{2} \rightarrow \boxed{i = 10\sqrt{2}}$$

$$F_2 = 10\sqrt{2}i - 10\sqrt{2}j$$

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2) Let $g(x) = 4 - x^2$.a) Find the average rate of change from -1 to 3 .

$$\frac{f(b) - f(a)}{b - a}$$

$$f(b) = 4 - 3^2 = 4 - 9 = -5$$

$$f(a) = 4 - (-1)^2 = 4 - 1 = 3$$

$$\frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = -2$$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

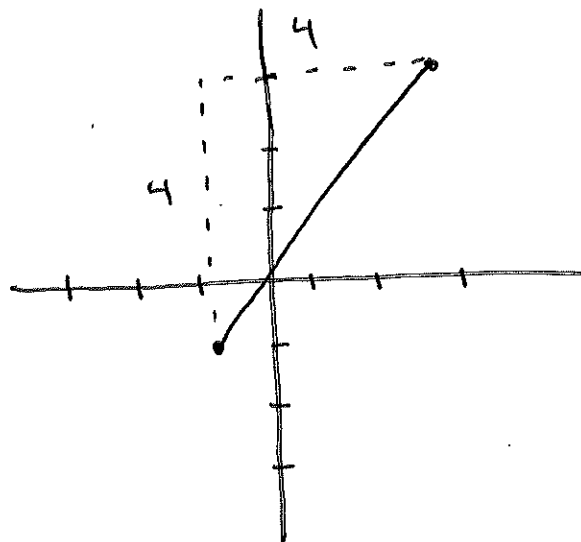
slope
↓

$$y - y_1 = m(x - x_1)$$

$$y - 3 = 1(x - 3)$$

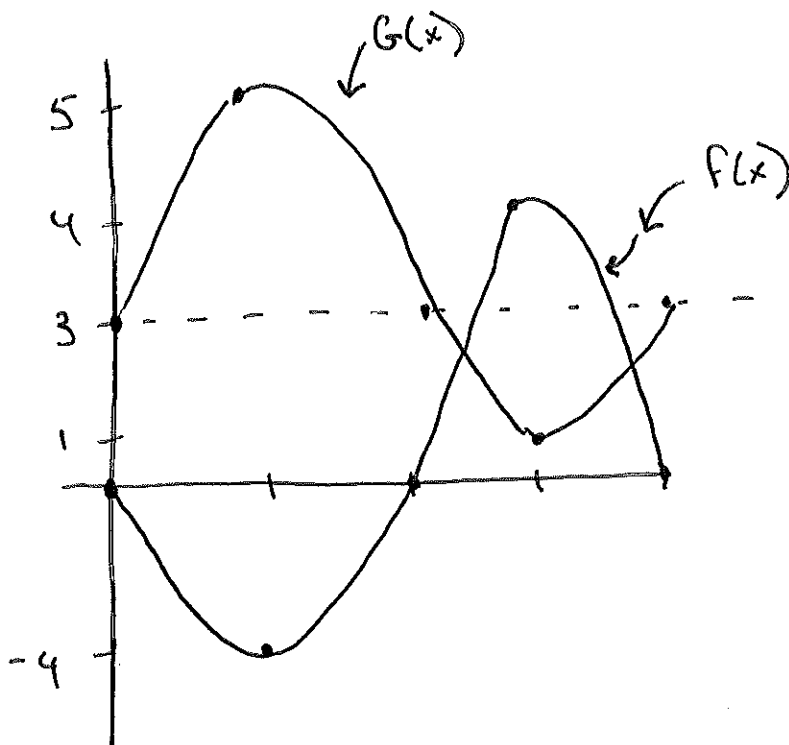
$$y + 3 = 1(x + 3)$$

$$y = 1x$$



3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$$\begin{aligned} -4\sin x &= 2\sin x + 3 \\ -2\sin x & \end{aligned}$$

$$\frac{-6\sin x}{-6} = \frac{3}{-6}$$

$$\sin x = -\frac{3}{6}$$

$$\sin x = -\frac{1}{2}$$

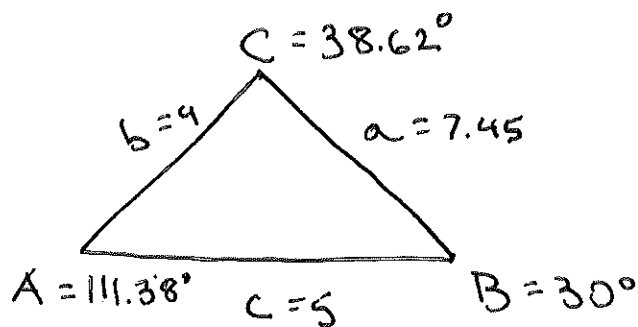
$$\sin x = -\frac{1}{2} \text{ at } \frac{7\pi}{6} \text{ and } \frac{11\pi}{6}$$

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4) Two sides and an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30^\circ$$



$b = 4$ SSA
 $c = 5$
 $B = 30^\circ$
 ↓
 possibility of two triangles

Triangle 1	Triangle 2
$A^\circ = 111.38$	$C^\circ = 141.38$
$C^\circ = 38.62$	$A^\circ = 8.62$
$a = 7.45$	$a = 1.12$

$$\frac{\sin 30^\circ}{4} = \frac{\sin C}{5} \rightarrow$$

$$\frac{5 \sin 30^\circ}{4} = \frac{\sin C}{1}$$

$$180 - 38.62 - 30 = A^\circ$$

$$A^\circ = \boxed{111.38}$$

$$\frac{\sin 30^\circ}{4} = \frac{\sin 111.38}{a}$$

$$a \sin 30^\circ = 4 \sin 111.38$$

$$a = \frac{4 \sin 111.38}{\sin 30^\circ} \rightarrow \frac{1}{\frac{1}{2}}$$

$$a = \frac{3.72473}{\frac{1}{2}} \rightarrow \boxed{a = 7.4494}$$

$$a = \frac{4 \sin 8.62}{\sin 30^\circ} \rightarrow \frac{0.59952}{\frac{1}{2}}$$

$$\boxed{a = 1.19904}$$

$$\sin C = \frac{5 \sin 30^\circ}{4}$$

$$\sin C = \frac{2.5}{4} \quad \sin C = 0.625$$

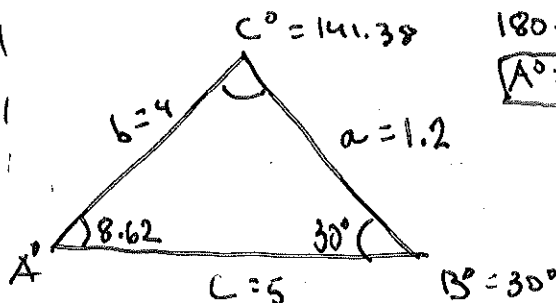
$$C^\circ = \sin^{-1}(0.625)$$

$$\boxed{C^\circ = 38.68}$$

$$\text{Triangle 2} \quad 180^\circ - 38.62 = \boxed{141.38 = C^\circ}$$

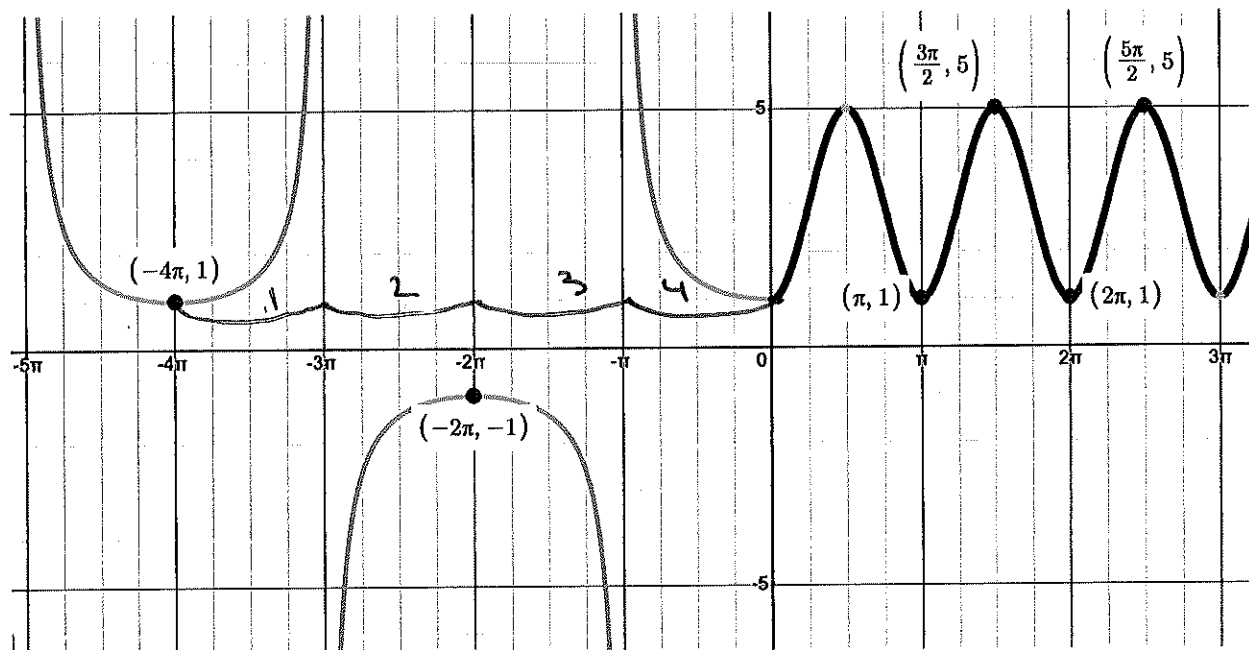
$$180 - 141.38 - 30 = A^\circ$$

$$\boxed{A^\circ = 8.62}$$



$$\frac{\sin 30^\circ}{4} = \frac{\sin 8.62}{a} \rightarrow \sin 30^\circ = \frac{4 \sin 8.62}{a}$$

5) Given the graph below find the function $f(x)$ and state the domain and range.



for red graph

$$\text{Domain} = (\mathbb{R}, \neq 2\pi k + 1) \cup [0, \infty)$$

$$\text{Range} = (-\infty, -1] \cup [1, \infty)$$

$$\text{Period: } \frac{2\pi}{B} = \frac{4\pi}{1} \rightarrow \frac{2\pi}{4\pi} = \frac{4\pi B}{4\pi} \rightarrow B = \frac{1}{2}$$

So, it's $\sec(\frac{1}{2}x)$

for black graph $\rightarrow$ So, $2\sin(2x) + 3$

$$\text{amplitude} = \frac{5-1}{2} = \frac{4}{2} = 2$$

$$\text{midline} = \frac{5+1}{2} = \frac{6}{2} = 3 \text{ (vertical shift)}$$

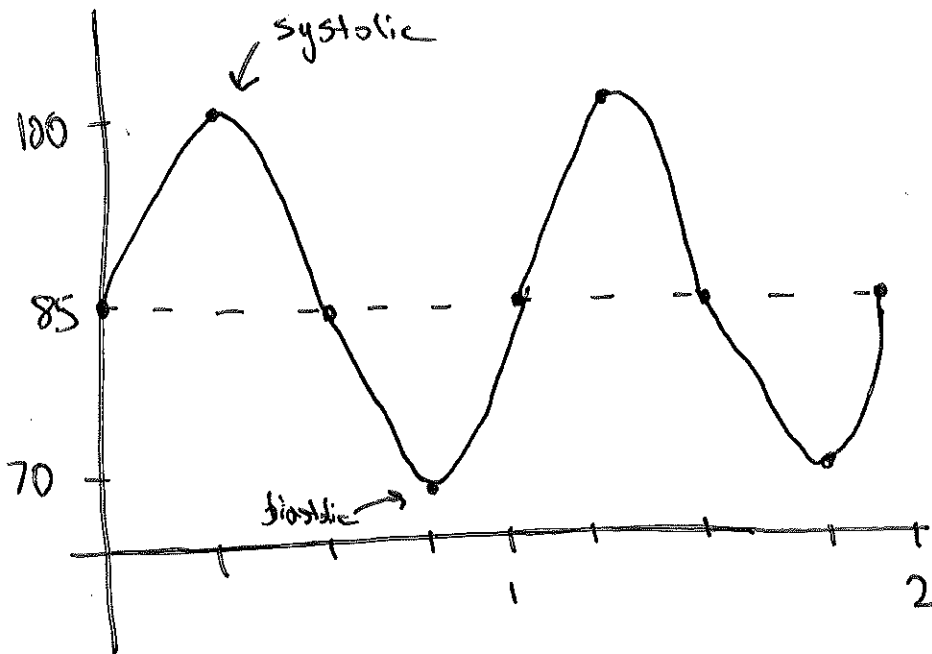
$$\text{Period is } \frac{5\pi}{2} - \frac{3\pi}{2} = \frac{2\pi}{2} = \pi \rightarrow \frac{2\pi}{B} = \frac{\pi}{1} = \frac{\pi B}{\pi} = \frac{2\pi}{\pi} = 2$$

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6) Blood pressure is modeled by the function $P(t) = 15 \sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$$\frac{2\pi}{2\pi} = 1 \leftarrow \text{period}$$

- b) The systolic pressure is 100
- c) The diastolic pressure is 70
- d) The heart rate is 60 beats per minute

$$\text{heart rate is } \frac{1}{T} \rightarrow \frac{1}{1}$$

$$\frac{1}{1} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = \frac{60}{1} = 60 \text{ bpm}$$